Term 2510
CDS6224 Statistical Data Analysis
Assignment 2 (30%)

This assignment comprises two case studies. Use R to make the analysis and graphics required for these case studies. The datasets are provided in two separate .csv files.

Case Study 1 (20 Marks)

Meet Kevin, the Production Manager at VoltPlus Technologies, a mid-sized company known for its reliable, long-lasting batteries. For the past decade, their flagship product—VoltCell Ultra—has proudly carried the tagline: "VoltCell Ultra – Powering Your World for 600 Hours and Beyond!"

This impressive claim wasn't just a marketing gimmick. Backed by years of engineering and statistical validation, the mean lifetime of these batteries consistently hovered around 600 hours. Customers loved the reliability, and the brand flourished.

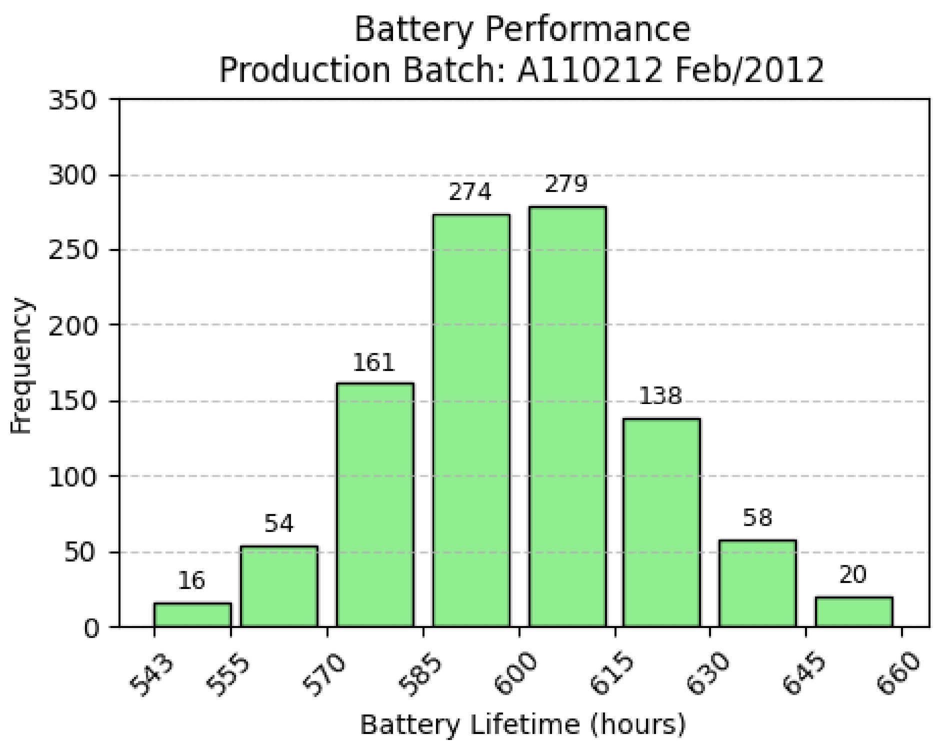
But times are changing.

A new production line was recently launched to cut costs and boost output. Now, after rolling out 1000 units from this new line, Kevin begins to worry. Early performance reports suggest that these new batteries might not live up to the VoltPlus legacy.

Kevin decides it's time to find out the truth. Is the new batch truly underperforming, or is it just a perception? As a budding data analyst, you're tasked with helping Kevin make a data- driven decision before he presents the findings to management. **Use R for your analysis.**

Part A – Descriptive Statistics [5 Marks]

1. Luckily, Kevin has found a histogram picture from one of the old reports in his office. However, no data was available for it. Using the histogram image, **estimate** the **mean** and **standard deviation** of the old battery batch using the frequency bar heights and bin ranges. [1.5+1.5 marks]



Mean

```
> # Define the frequencies and class midpoints
> midpoints <- c(549, 562.5, 577.5, 592.5, 607.5, 622.5, 637.5, 652.5)
> frequencies <- c(16, 54, 161, 274, 279, 138, 58, 20)
>
> # Multiply midpoints by their corresponding frequencies
> products <- midpoints * frequencies
>
> # Calculate the estimated mean
> estimated_mean <- sum(products) / 1000
>
> # Print the result
> print(estimated_mean)
[1] 599.904
```

Standard Deviation

```
> # Given data
> midpoints <- c(549, 562.5, 577.5, 592.5, 607.5, 622.5, 637.5, 652.5)
> frequencies <- c(16, 54, 161, 274, 279, 138, 58, 20)
> n <- 1000
>
> # Estimate mean
> mean_est <- sum(midpoints * frequencies) / n
>
> # Estimate variance
> squared_diff <- (midpoints - mean_est)^2
> weighted_squared_diff <- squared_diff * frequencies
> variance_est <- sum(weighted_squared_diff) / n
>
> # Standard deviation
> sd_est <- sqrt(variance_est)
>
> # Output
> print(sd_est)
[1] 20.89753
```

2. From the raw data for new batch of battery (given in csv file "BatteryLifeNew.csv"):
   a. Calculate the **sample mean** and **sample standard deviation**. [0.5+0.5 marks]

```
> #Load the CSV file
> data <- read.csv("BatteryLifeNew.csv")
>
> Battery_Life <-data$LifetimeHrs
>
> #Calculate mean and standard deviation
> mean_battery <- mean(Battery_Life)
> sd_battery <- sd(Battery_Life)
>
> #Print the results
> cat("Sample Mean:", mean_battery, "\n")
Sample Mean: 580.3862
> cat("Sample Standard Deviation:", sd_battery, "\n")
Sample Standard Deviation: 19.58436
```

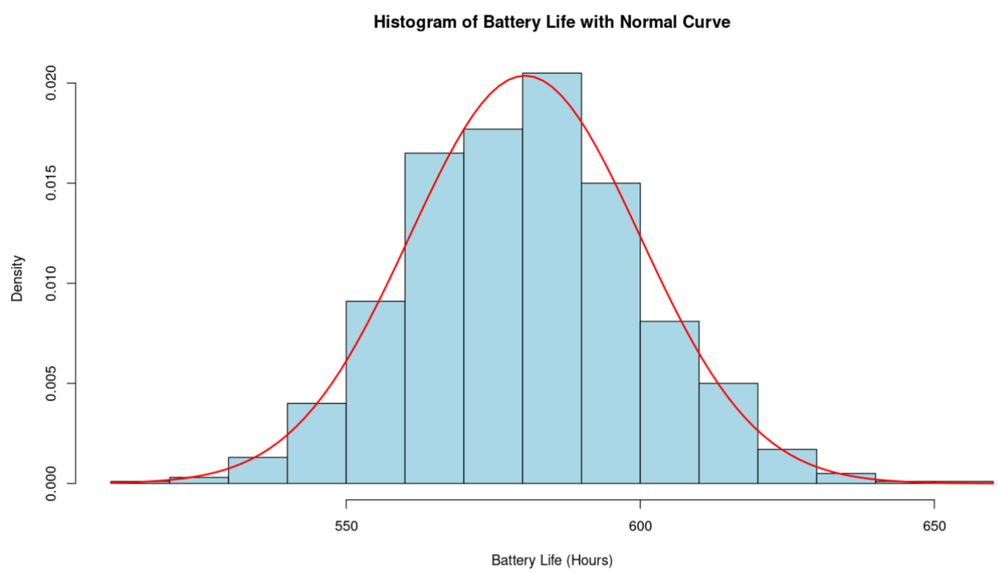
   b. Compute the **95% confidence interval** for the mean (CI_upper, and CI_lower). (Hint: Use t-tests) [0.5+0.5 mark]

```
> #Compute 95% Confidence Interval using t-test
> t_test_result <- t.test(Battery_Life, conf.level = 0.95)
>
> # Extract the confidence interval
> ci_lower <- t_test_result$conf.int[1]
> ci_upper <- t_test_result$conf.int[2]
>
> # Print the confidence interval
> cat("Lower Bound:", ci_lower, "\n")
Lower Bound: 579.1709
> cat("Upper Bound:", ci_upper)
Upper Bound: 581.6015
```

Part B – Visualization [4 Marks]

1. Create a histogram (with appropriate Y-axis) of the new batch and draw an overlay normal curve. [1+1 marks]

```
> Battery_Life <- data$LifetimeHrs
> ID <- data$BatteryID
>
> # Histogram with overlay normal curve
> hist(Battery_Life,
+      breaks = 20,
+      freq = FALSE,  # Use density for Y-axis
+      main = "Histogram of Battery Life with Normal Curve",
+      xlab = "Battery Life (Hours)",
+      col = "lightblue",
+      border = "black")
>
> # Add a normal curve
> curve(dnorm(x, mean = mean(Battery_Life), sd = sd(Battery_Life)),
+       col = "red", lwd = 2, add = TRUE)
```




```
> Battery_Life <- data$LifetimeHrs
> ID <- data$BatteryID
>
> # Histogram with overlay normal curve
> hist(Battery_Life,
+     breaks = 20,
+     freq = FALSE,  # Use density for Y-axis
+     main = "Histogram of Battery Life with Normal Curve",
+     xlab = "Battery Life (Hours)",
+     col = "lightblue",
+     border = "black")
>
> # Add a normal curve
```

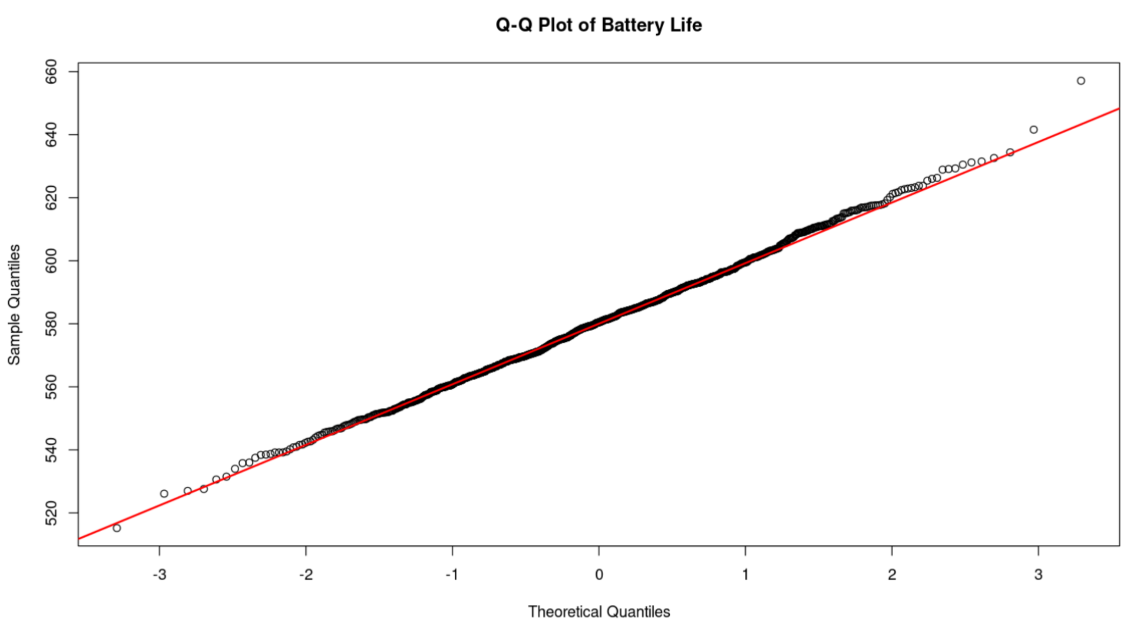
```
> curve(dnorm(x, mean = mean(Battery_Life), sd = sd(Battery_Life)),
+       col = "red", lwd = 2, add = TRUE)
```

2. Generate a Q-Q plot for the new data to assess normality (Use Shapiro-Wilk Statistic). Make a comment on the results you get. [1+1 marks]

```
> # Q-Q Plot
> qqnorm(Battery_Life, main = "Q-Q Plot of Battery Life")
> qqline(Battery_Life, col = "red", lwd = 2)
>
> # Shapiro-Wilk Test for Normality
> shapiro_result <- shapiro.test(Battery_Life)
> print(shapiro_result)

        Shapiro-Wilk normality test

data:  Battery_Life
W = 0.9986, p-value = 0.6243
```




```
> # Q-Q Plot
> qqnorm(Battery_Life, main = "Q-Q Plot of Battery Life")
> qqline(Battery_Life, col = "red", lwd = 2)
>
> # Shapiro-Wilk Test for Normality
> shapiro_result <- shapiro.test(Battery_Life)
> print(shapiro_result)
```

Interpretation:
The Q-Q plot shows that the data is approximately normal as most of the points

fall on the red line.The Shapiro-Wilk test gives a p-value= 0.6243, that is p-value > 0.05, so there is sufficient evidence to prove that it is normally distributed.

Part C – Hypothesis Testing [11 Marks]

1. State the null (H0) and alternative (H1) hypotheses. (Hint: Null hypothesis is the default scenario) [1 mark]

   Is the mean lifetime of new batteries equal to 600 hours?

   $H_0: \mu = 600$

   $H_1: \mu < 600$

2. Conduct a one-sample t-test. Provide the t-statistic, degrees of freedom, p-value. Use α = 0.05 to state if H0 should be rejected. [3+3 marks]

```
> # Perform one-sample t-test comparing new batch to old batch mean
> t_result <- t.test(Battery_Life, mu = 600)
>
> # View results
> print(t_result)

        One Sample t-test

data:  Battery_Life
t = -31.67, df = 999, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 600
95 percent confidence interval:
 579.1709 581.6015
sample estimates:
mean of x
 580.3862
```

t-statistic = -31.67
degrees of freedom = 999
p-value < 2.2e-16

1. Interpret the results:
   a. Is H0 rejected or accepted? State.[1 mark]
      The p-value < 2.3e-16 which is less than α = 0.05. H_0 is rejected at α = 0.05.
   b. What do they suggest about the quality of the new batch?

Should VoltPlus continue with the new production line and be able to justify their tagline “VoltCell Ultra – Powering Your World for 600 Hours and Beyond!”? [3 marks]

The quality of the new batch is not as good as the old batch.

Based on the result, the sample mean of the new batch is lower than 600 hours. The p-value also provides sufficient evidence to prove that the mean lifetime of new batteries equals 600 hours.

So, VoltPlus should not continue to use the new production line and be able to justify their tagline “VoltCell Ultra – Powering Your World for 600 Hours and Beyond!”.

Case Study 2 (10 Marks)

FreshFizz is a popular fast food outlet that introduced a new drink flavor, **Mango Zing**, in
June 2024. FreshFizz is analyzing the effect of the introduction of the new Mango Zing flavor on sales. The company wants to understand if the new flavor has caused a significant change in overall sales and whether the sales of existing drinks (like Cola Classic, Lemon Light, and Berry Blast) show any significant differences before and after Mango Zing was introduced. **Use R for your analysis.**

You are provided with the dataset **"FreshFizz.csv"** for 2024, which includes daily sales for drinks and food. The goal is to use statistical methods like ANOVA and correlation analysis to examine these changes. You are given a dataset containing:

- Date: Daily date in YYYY-MM-DD format (R-compatible)
- Cola_Classic: Units of Cola Classic sold that day
- Lemon_Light: Units of Lemon Light sold that day
- Berry_Blast: Units of Berry Blast sold that day
- Mango_Zing: Units of Mango Zing sold (0 before June 1, >0 after)
- Unit_Price: Unit price for drinks (same for all flavours)
- Sandwiches_Sold: Units of sandwiches sold that day
- Chips_Sold: Units of chips sold that day
- Cookies_Sold: Units of cookies sold that day
- Sandwich_Price: Price of one sandwich
- Chips_Price: Price of one pack of chips
- Cookies_Price: Price of one pack of cookies
- Total_Sales: Total revenue for the day (drinks + food

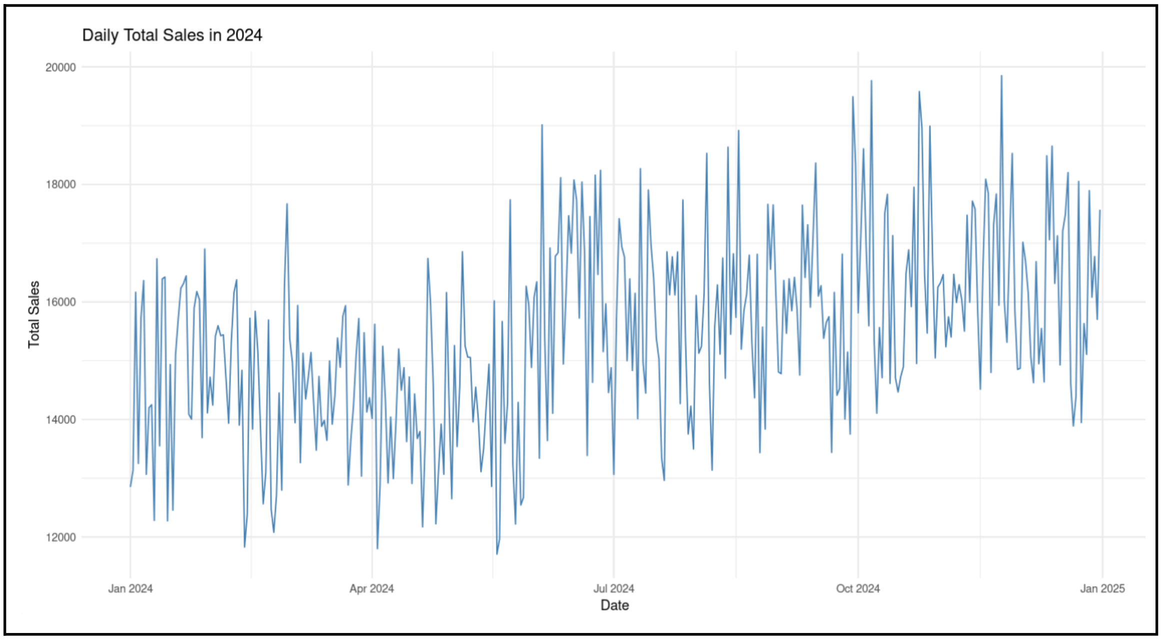
combined) Note: Mango Zing is only present from **June 2024**

onward.

Part A: Plotting and Summary Statistics (3 marks)

1. Plot the daily total sales for the entire year using a suitable chart. (1 mark)

```
> # Load the dataset
> data <- read.csv("FreshFizz.csv")
>
> # Convert Date to proper format
> data$Date <- as.Date(data$Date)
>
> # Plot daily total sales using line plot
> library(ggplot2)
```

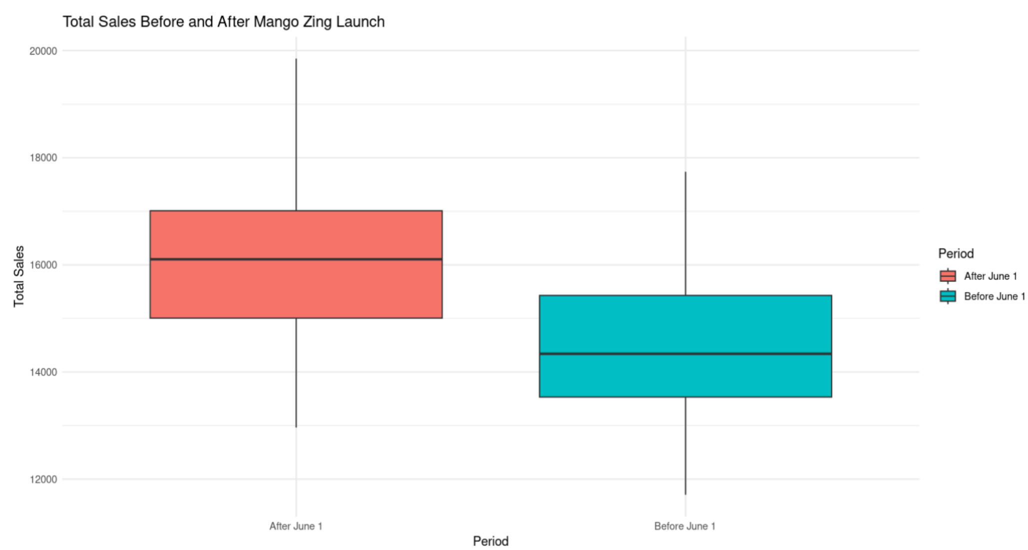


2. Plot a side-by-side box plot of daily sales, and calculate and summarize the mean and standard deviation of **daily total sales** for the following periods:
   a. Before June 1 (pre-launch of Mango Zing)
   b. After June 1 (post-launch of Mango Zing) (1+0.5+0.5 marks)

```
> # Define cutoff date
> cutoff_date <- as.Date("2024-06-01")
>
> # Create subsets
> pre_june <- subset(data, Date < cutoff_date)
> post_june <- subset(data, Date >= cutoff_date)
>
>
> # Add a new column for period
> data$Period <- ifelse(data$Date < cutoff_date, "Before June 1", "After June 1")
>
> # Boxplot
> ggplot(data, aes(x = Period, y = Total_Sales, fill = Period)) +
+   geom_boxplot() +
+   labs(title = "Total Sales Before and After Mango Zing Launch",
+        x = "Period",
+        y = "Total Sales") +
+   theme_minimal()
```

```
> # Summary statistics
> mean_pre <- mean(pre_june$Total_Sales)
> sd_pre <- sd(pre_june$Total_Sales)
>
> mean_post <- mean(post_june$Total_Sales)
> sd_post <- sd(post_june$Total_Sales)
>
> # Print results
> cat("Before June 1:\n")
Before June 1:
> cat("Mean Total Sales:", round(mean_pre, 2), "\n")
Mean Total Sales: 14432.7
> cat("Standard Deviation:", round(sd_pre, 2), "\n\n")
Standard Deviation: 1336.8

> cat("After June 1:\n")
After June 1:
> cat("Mean Total Sales:", round(mean_post, 2), "\n")
Mean Total Sales: 16114.2
> cat("Standard Deviation:", round(sd_post, 2), "\n")
Standard Deviation: 1459.95
```



Part B: One-Way ANOVA for Sales Differences Before and After Mango Zing (1 mark)

Perform a One-Way ANOVA to test whether there is a significant difference in **total sales** before and after the introduction of Mango Zing (using the Date column to separate before and after June 1). State your hypothesis, p-value, and conclusion.

Hypothesis:
Null hypothesis H_0:
There is no significant difference in daily total sales before and after the introduction of Mango Zing.
Null hypothesis H_1:
There is a significant difference in daily total sales before and after the introduction of Mango Zing.

```
> # Create a period column
> data$Period <- ifelse(data$Date < as.Date("2024-06-01"), "Before", "After")
>
> # Run ANOVA
> anova_result <- aov(Total_Sales ~ Period, data = data)
>
> # Show result
> summary(anova_result)
             Df    Sum Sq   Mean Sq F value Pr(>F)
Period        1 251286383 251286383   126.4 <2e-16 ***
Residuals   364 723841014   1988574
```

p-value $< 2 \times 10^{-16}$

Since p-value is < 0.05, null hypothesis H_0 is rejected.

Conclusion:
Thus, there is a significant difference in daily total sales before and after the introduction of Mango Zing. This suggests that the launch of Mango Zing had a positive impact on overall sales.

Part C: ANOVA to Test Cannibalization of Other Flavors (Post-Launch) (1.5 marks)

After the launch of Mango Zing on June 1, there's concern that it may have reduced the sales of the existing three flavors — Cola Classic, Lemon Light, and Berry Blast.

Perform a One-Way ANOVA to compare the mean sales of these three original flavors in the post-launch period only (from June 1 onward). State the hypothesis and interpret the result. Comment on whether Mango Zing may have cannibalized/decreased the sales of other flavors. (0.5+0.5+0.5 marks)

Hypothesis:
Null hypothesis H_0: $\mu_{Cola\ Classic} = \mu_{Lemon\ Light} = \mu_{Berry\ Blast}$
The mean sales of Cola Classic, Lemon Light and Berry Blast are equal in the post-launch period.

Null hypothesis H_1:
At least one of the original flavors has a different mean sales level compared to the others post-launch.

```
> # Filter only post-launch data
> post_launch <- subset(data, Date >= as.Date("2024-06-01"))
>
> # Reshape to long format for ANOVA
> library(tidyr)
> long_data <- pivot_longer(post_launch,
+                           cols = c(Cola_Classic, Lemon_Light, Berry_Blast),
+                           names_to = "Flavor",
+                           values_to = "Units_Sold")
>
> # Run one-way ANOVA
> anova_flavors <- aov(Units_Sold ~ Flavor, data = long_data)
>
> # Show ANOVA result
> summary(anova_flavors)
             Df   Sum Sq Mean Sq F value Pr(>F)
Flavor        2 10786861 5393431   103.4 <2e-16 ***
Residuals   639 33342205   52179
```

Interpretation:
p-value $< 2 \times 10^{-16}$

Since p-value is < 0.05, null hypothesis H_0 is rejected.

Conclusion:
There is a significant difference in the mean sales of Cola Classic, Lemon Light and Berry Blast after the launch of Mango Zing. This means that Mango Zing may harm sales of at least one of the original flavors by taking customers away and changing the way sales are spread out among the other drinks.

Part D: Correlation Analysis Between Mango Zing and Total Sales (2 marks)

Determine the relevant correlation coefficient between the launch of Mango Zing and the daily total revenue. Explain whether the correlation is positive or negative, and whether it is strong or weak. Conclude whether the new flavor had a noticeable impact on overall sales.

(Hint: Work out the appropriate correlation method and variables to be used. Do you need to construct a new compound variable? Or, one hot encoding?)

```
> # Create binary variable for Mango Zing period
> #0 for dates before June 1, 2024
> #1 for dates on or after June 1, 2024
> data$Mango_Zing_Launched <- ifelse(data$Date >= as.Date("2024-06-01"), 1, 0)
>
> # Compute Pearson correlation between Mango Zing launch and Total_Sales
> correlation <- cor(data$Mango_Zing_Launched, data$Total_Sales, method = "pearson")
>
> # Print correlation coefficient
> cat("Correlation coefficient:", round(correlation, 3))
Correlation coefficient: 0.508
```

Interpretation:
Correlation coefficient =0.508, the correlation coefficient is positive. This means that when Mango Zing is available, the total sales tend to increase.
The correlation coefficient is a moderate positive correlation between the launch of Mango Zing and the daily total revenue.

The correlation coefficient is a moderate positive correlation between the launch of Mango Zing and the daily total revenue. This indicates that Mango Zing has a beneficial effect on total sales but it is not the only factor. The relationship is not strong enough to say that Mango Zing alone controls sales trends.

Part E: Should Mango Zing Be Continued? (2.5 marks)

Based on the results of the ANOVA and correlation analyses, as well as the overall trends in sales before and after the introduction of Mango Zing, comment on whether Mango Zing should be continued as a product offering. Your answer should consider:

- The significance of Mango Zing's sales (does it contribute significantly to overall sales?).
- The impact of Mango Zing on other flavors' sales (is the decline in sales of other flavors offset by Mango Zing?).
- Overall revenue trends (did total sales increase after the launch of Mango Zing?).

Support your conclusion with relevant findings from the analyses (ANOVA p-values, correlation values, and general sales trends).

The mean total sales before June 1 is 14432.70, while the mean total sales after June 1 is 16114.20. This indicated that the mean total sales increased by approximately

11.6% post-launch. The standard deviation before and after launching of Mango Zing also increased from 1336.8 to 1459.95. A One-Way ANONA on total sales before and after the latch of Mango Zing shows that the p-value $< 2 \times 10^{-16}$. This means that total sales increased significantly post-launch.
A post-launch ANONA compare Cola Classic, Lemon Light and Berry Blast shows that p-value $< 2 \times 10^{-16}$ and F-value =104.4. The overall sales increase suggests that any decline is outweighed by gains from Mango Zing when the cannibalization of one or more flavors.
The correlation coefficient is 0.508, which indicates a moderate positive correlation between the launch of Mango Zing and the daily total revenue. This shows that Mango Zing contributes positively even if it is not the only factor.
Thus, VoltPlus Technologies should continue offering Mango Zing. It enhances total sales and appears to attract or retain customers.

Submission Guidelines:

Submission Format: Submit your assignment as a **PDF** file only.

Your PDF must include:

- Well-commented R scripts
- All relevant R outputs (e.g., tables, plots, and results)
- **Clear written answers and interpretations for each section**

File Naming Format:

Name your file using the format:

LectureSection_Name_StudentID_CDS6224_Assignment2.pdf

Example:

TC1L_MdZubair_1020304050_CDS6224_Assignment2.pdf

Please ensure all components are complete and properly formatted before submitting.

Submission Details:

22-June-2025 (Sunday), 23:59 Hrs.

eBWise.

Individual Submission.